

PROJECT : OMNICHANNEL RETAIL & LOCATION INTELLIGENCE

Candidate: Natthaorn Lapprasitsuk

Target Role: Senior Data Analyst / Data Analytics Specialist

Tech Stack: Python (Pandas, NumPy, Scikit-Learn), SQL, Tableau

1. Executive Summary & Business Challenge

This project addresses an omnichannel retail optimization challenge by analyzing **1,000 transactions** across three distinct sales channels: **Offline Stores, Online Websites, and Roboshops**. The primary objective is to maximize regional profitability by identifying cost inefficiencies and establishing a data-driven framework for future store location selection.

2. Data Infrastructure & ETL Pipeline (Python)

- **Data Aggregation:** Built a Python-based ETL pipeline on macOS to extract and clean raw sales transaction data alongside separate regional monthly expense datasets (leasing, marketing, and operations).
- **Data Treating:** Handled missing revenue values using channel-specific median imputation to ensure data integrity before calculating core retail metrics: **Revenue, Sales Volume, and Average Order Value (AOV)**.

3. Commercial & Cost Analytics Dashboard (Tableau)

The treated dataset was processed into an executive dashboard to isolate profit margin leakages and evaluate resource efficiency across regions.

Omnichannel Retail Strategy & Analytics

Total Revenue

780,045

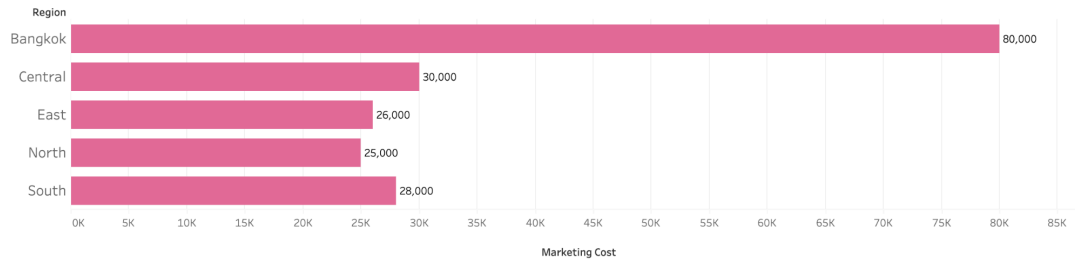
Average Order Value (AOV)

361,887

Profit Margin (Pct)

Region	Profit Margin (Pct)
Bangkok	-116.1
Central	27.97
East	37.88
North	24.81
South	30.77

Marketing Cost by Region

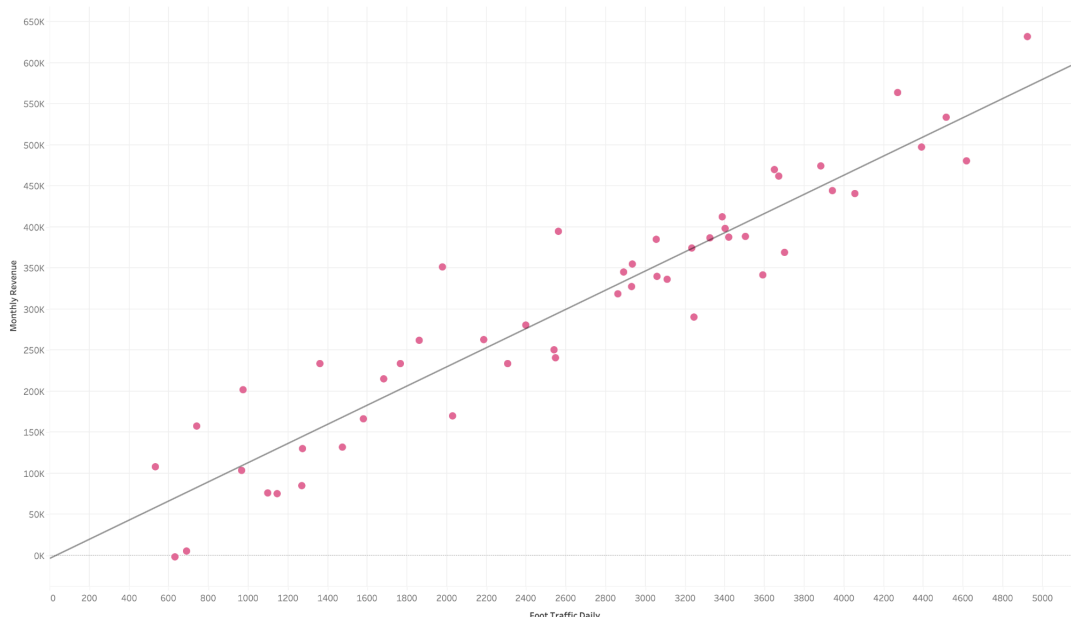


- **Key Insight:** While the **Bangkok and Central regions** remain the primary revenue engines, Bangkok experiences severe margin compression due to disproportionately high leasing and operational overhead. Conversely, the **East and South regions** demonstrate the healthiest and most efficient profit margins (ranging from 30% to 37%).

4. Advanced Statistical Modeling & Site Selection

To eliminate guesswork from future real estate expansions, a **Linear Regression Model** was developed using a synthetic dataset of 50 potential store locations to isolate the true drivers of monthly revenue.

Relationship between Food Traffic Daily & Monthly Revenue



- **Statistical Findings:**

- **Daily Foot Traffic** was identified as the strongest, highly significant positive predictor of revenue.
- **Competitor Density** exhibits a quantifiable negative correlation, serving as a critical constraint for market saturation within a 1-km radius.

5. Business Action Plan & ROI Impact

1. **Cost Optimization:** Implement an immediate **15% marketing budget reallocating** from low-margin physical areas into high-conversion digital acquisition channels.
2. **Asset Transition:** Transition underperforming, high-overhead physical retail spots into leaner, lower-overhead **Roboshops** to protect regional net profit margins.
3. **Expansion Framework:** Enforce a strict **Data-Driven Site Selection Framework** for Q3 store rollouts—prioritizing high-foot-traffic zones *only* when competitor density falls below the critical market threshold.